Dulitha Prasanna Kulathunga

Postdoctoral Research Associate

dulithaprasanna@gmail.com Chapel Hill, NC dulithaprasanna.github.io dulithaprasanna

Summary

- Computational and theoretical chemist and biophysicist working at the interface of molecular simulation, machine learning, and scientific discovery. I combine molecular dynamics, quantum chemistry, and enhanced-sampling simulations with active machine learning and data-driven optimization to find promising molecules, reaction conditions, and protein sequences with as few costly experiments or simulations as possible.

Education

Iowa State University & Ames National Laboratory, Physical Chemistry — Ames, IA
- Thesis advisors: Prof. Theresa L. Windus and Prof. Davit Potoyan. — 2021 – Apr 2026
- Dissertation: *Exploration without Exhaustion*, defended April 10, 2026.

University of Colombo, Computational Chemistry — Colombo, Sri Lanka
- Completed 2020. Advisor: Prof. Samantha Weerasinghe.
- CUFSAA Gold Medal for Best Performance in Computational Chemistry (2019).

University of Colombo School of Computing (UCSC), Information Technology — Colombo, Sri Lanka
- Completed 2019.

Experience

University of North Carolina at Chapel Hill, Postdoctoral Research Associate, Miao Lab — Department of Pharmacology, Chapel Hill, NC
- Advisor: Prof. Yinglong Miao. — 2026 – present
- Gaussian accelerated molecular dynamics (GaMD) enhanced-sampling simulations of RNA and RNA–ligand systems. — 1 year
- Developing AI-driven models for RNA-targeted drug discovery and virtual screening.

Iowa State University & Ames National Laboratory, Graduate Research Assistant — Ames, IA
- **AMLRO:** built an open-source active learning framework for single- and multi-objective reaction optimization, with a Flask/Streamlit GUI for open-loop use and SHAP-based explainability. Applied it to Paal-Knorr flow synthesis, closed-loop diglycolamide (DGA) flow synthesis, and aldehyde-selective Wacker oxidation in a 7-dimensional parameter space. — 2021 – Apr 2026
- **CyRxnOpt:** co-developed a unified interface for benchmarking reaction optimization algorithms, integrated into the RxnRover automation ecosystem; published in *Digital Discovery* (2026) as co-first author. — 5 years
- **ALSEBO:** developed a coevolution-informed Bayesian optimization framework over a generative latent sequence landscape for data-efficient protein design; designed novel green fluorescent protein variants validated by MD simulations.
- **Enzyme thermal adaptation:** studied serine proteases and cold-adapted β-glucosidases from psychrophilic to thermophilic organisms using molecular dynamics, bioinformatics, and unsupervised machine learning.
- Collaborated with Oak Ridge National Laboratory and the Critical Materials Innovation Hub (CMI) on multi-objective optimization of molten salt electrochemical graphitization.

Iowa State University, Graduate Teaching Assistant — Ames, IA
- General chemistry laboratories and recitations (Chem 163/163L, 167L, 177L). — 2021 – 2022, 1 year

University of Colombo, Teaching Assistant — Colombo, Sri Lanka
- Computational chemistry laboratory courses for undergraduate and M.Sc. students. — 2020 – 2021, 1 year

University of Colombo, Undergraduate Research Assistant — Colombo, Sri Lanka
- Compared all-atom (AMBER99SB-ILDN) and united-atom (KBFF) force fields for interactions between β-amyloid and model amino acids with unsaturated side chains, using GROMACS. — 2019 – 2020, 1 year

Open Source Projects

AMLRO (Active Machine Learning Reaction Optimizer)
Active learning framework for single- and multi-objective optimization of chemical reaction conditions.
- Closed-loop (automated) and open-loop (human-in-the-loop) optimization, with a web GUI.
- Explainable AI with SHAP to rank the influence of reaction variables.
- Released as DOE software: doi:10.11578/dc.20260205.1.

AMLRO GUI
Web interface for running AMLRO active learning optimization campaigns without writing Python (Flask API backend, React/TypeScript frontend).
- Released as DOE software: doi:10.11578/dc.20260928.1.

ALSEBO (Active Learning Sequence Exploration via Bayesian Optimization)
Generative protein design by Bayesian optimization in a learned latent space.
- Variational autoencoder (VAE) latent space for protein sequences.
- Coevolution-informed Bayesian optimization to propose new variants with few measurements.
- Demonstrated on avGFP fluorescence (bioRxiv 2026).

CyRxnOpt
Common Python interface for reaction optimization algorithms, usable from a CLI or inside self-optimizing platforms.
- Published in *Digital Discovery* (2026): doi:10.1039/D6DD00180G.

Presentations

2026: Ph.D. Final Defense: "Exploration without Exhaustion", Department of Chemistry, Iowa State University & Ames National Laboratory.

2026: Poster: "ALSEBO: Bayesian-Guided Exploration of the Green Fluorescent Protein Latent Generative Landscape", Chemistry Department Open House, Iowa State University, Ames, IA.

2025: Oral: "From data to discovery: efficient reaction design with the active learning multi-objective reaction optimizer (AMLRO) framework", ACS Fall 2025, Washington, D.C.

2025: Oral: "AI/ML Tools for Chemical Process Optimization: Trial-and-Error to Data-Driven Discovery", Critical Materials Innovation Hub (CMI) FA4 Meeting.

2025: Poster: "Molecular insights into cold-adapted beta-glucosidases", Biophysical Society 69th Annual Meeting.

2024: Poster: "AI-Driven Optimization for Chemical Synthesis", Chemistry Department Open House, Iowa State University, Ames, IA.

2024: Poster: "Evolutionary thermal adaptations of enzymes at temperature extremes", Biophysical Society 68th Annual Meeting.

2023: Preliminary Oral Examination: "Computational Design of Enzymes and Chemical Reactions", Department of Chemistry, Iowa State University.

2023: Oral: "Implementing a unified object-oriented framework for benchmarking chemical reaction optimization algorithms", ACS Midwest and Great Lakes Joint Regional Meeting.

2023: Poster: "Thermal adaptations of co-evolved psychrophilic, mesophilic, and thermophilic enzymes", Biophysical Society 67th Annual Meeting.

2020: Oral: "Developing a Machine Learning Web Application for Predicting Liver Patients", APACPH 2020 Conference: Public Health in the New Normal.

2020: Poster: "Comparison of structural properties and energetics of model amino acids containing unsaturated side chains using all-atom and united-atom force fields in MD simulations", SLAAS 76th Annual Scientific Sessions.

Awards

Ames National Laboratory Innovation Incentive Award (for CyRxnOpt) 2026
Ames National Laboratory

Klaus Ruedenberg Theoretical Chemistry Award 2025
Department of Chemistry, Iowa State University

Drtina Chemistry Graduate Student Travel Award 2025
Received in 2024 and 2025.
Department of Chemistry, Iowa State University

Edward V. Sayre Chemistry Scholarship 2024
Department of Chemistry, Iowa State University

CUFSAA Gold Medal for Best Performance in Computational Chemistry 2019
University of Colombo

Skills

Programming & Apps

Machine Learning

Molecular Simulation

Computational Chemistry & Cheminformatics